



Firefox adds 10 new extensions
Mozilla has added support for ten new extensions in the latest Firefox Nightly for Android. <http://dgit.in/oct20-59>



Flying into Hurricane Laura
Microsoft Flight Simulator players were flying into Hurricane Laura as it approached the US. <http://dgit.in/oct20-60>

Synology DiskStation DS920+ NAS

Snappy and feature-laden

The DS920+ is a direct upgrade to the DS918+ that won the Zero1 award last year. It was the top-of-the-line NAS for home consumers from Synology that comes laden with features that make file management seem like child's play. With the DS920+, Synology is also bringing forth a few key improvements to the hardware. Let's take a look at all that they've packed in.

Build Quality

The Synology DS920+ is a 4-bay NAS with a similar construction as the DS918+. It has the same as the internal chassis and the mainboard mount are the only metallic components and the rest of the unit is made up of plastic. The locking mechanisms and the HDD trays are still the same.

The rear panel on which the ports and fans are mounted happen to be made of metal as well. We do see the main SoC mounted in a fashion that makes it slightly isolated from the chamber housing the hard drives and there's even spacing to ensure airflow over the SoC heatsink. The DS920+ has a single 4G DDR4 RAM DIMM installed with space for another one. The DS918+ made it easier to access both RAM slots but the DS920+ only allows you to access one module from within the cage whereas the other one requires a bit of disassembly.

For the SoC, Synology is using the Intel Gemini Lake platform (Intel Atom J4125) which is a step up from the DS918+ and brings DDR4 support. The Intel Atom J4125 is a 4 core / 4 thread processor with Intel HD 600 Graphics featuring 12 Execution Units. It has a base clock of 2.0 GHz and can rise to 2.7 GHz. Gemini Lake also adds support for USB 3.1 and Type-C interfaces but Type-C as a storage interface is yet to be as widespread as Type-A connectors. Most external SSDs that we've come across ship with Type-C and



Type-A cables and that could changeover to just Type-C in the years to come. So hopefully, we'll see NAS units adopting Type-C around the same time.

Performance

We used four Seagate IronWold 2TB drives which were all CMR (Conventional Magnetic Recording) drives so there would be no problems related to rebuilding a RAID array if things had gone south. We threw in NVMe SSDs to help speed up the caching and a 240 GB SSD did just fine with the job. Over the course of one month, we had a cache hit rate of 48 per cent, on a weekly level it came down to 9 per cent and on some days, we'd hit 82 per cent. We'd configured the drives on a RAID 5 array since that seems to offer the best of data redundancy as well as performance. We could switch to RAID 0 but then again, we're using HDDs on Gigabit Ethernet, so it's not exactly prudent to give up redundancy to negligible speed gains.

In our RAID 5 benchmarks, the Synology DS920+ outperforms the DS918+ by a decent margin. In fact, we were hitting the Gigabit Ethernet threshold speeds in quite a few benchmarks and it seems that we might have to upgrade our networking infrastructure to get the little extra performance that we're

missing out on. Thankfully, the

DS920+ comes with two Gigabit Ethernet ports and you can team them together using Link Aggregation to push the envelope even further.

When streaming four HD videos off the NAS, we could see a throughput of about 114.89 MBps which is practically the highest we've seen so far. For copying a massive chunk of irregularly sized files off and to the NAS we saw an average throughput of 12.3 to 13.6 MBps which is again, the highest we've seen across NAS devices so far. The NAS was always quick to respond whenever accessed even after extended periods of inactivity. All in all, this is a snappy unit. All tests

were performed on DSM 6.2.3 which is the latest firmware for the device.

Bottomline

The Synology DS920+ is a decent upgrade to the DS918+. The move up to Intel Gemini Lake and DDR4 are welcome changes that add to the unit's responsiveness. Moreover, with the newer processor, Synology has plenty of performance headroom to introduce newer features and applications with AI components that can leverage the HD600 IGP for an improved user experience. At 60K, the DS920+ is one of the most feature-rich NAS devices you can get.

—Mithun Mohandas

SPECIFICATIONS

CPU: Intel Celeron J4125 Quad Core | RAM: 4 GB DDR4 | BAYS: 4 (+2 slots for M.2 SSDs) | HOT SWAP SUPPORT: Yes | I/O: eSATA, 2x USB 3.0 | LAN PORTS: 2x Gigabit | WEIGHT: 2.24 kg | FAN: 2x 92 mm | WARRANTY: 3 years.

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